

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = \int_0^{\infty} \int_0^{\infty} e^{-(x+y)} dx dy$$

$$= \int_0^{\infty} e^{-x} dx \times \int_0^{\infty} e^{-y} dy$$

$$= [-e^{-x}]_0^{\infty} \times [-e^{-y}]_0^{\infty} = 1$$

$$\frac{\int \int f(x,y) dx dy}{\int f(x) dx}$$

$$P(X > 1, Y > 1) = \int_1^{\infty} \int_1^{\infty} f(x,y) dx dy = \int_1^{\infty} \int_1^{\infty} e^{-(x+y)} dx dy =$$

• Let $f(x,y)$ be the joint PDF of the RV (X,Y) .

$$\text{If } B \in \mathbb{R}^2, P[(X,Y) \in B] = \iint_B f(x,y) dx dy \quad \checkmark$$

$$\text{The marginal pdf of } X \text{ is } \underline{f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy}$$

$$\text{The marginal pdf of } Y \text{ is } \underline{f_Y(y) = \int_{-\infty}^{\infty} f(x,y) dx}$$

The conditional pdf of X given $Y=y$ is

$$f_{X|Y=y}(x|y) = \frac{f(x,y)}{f_Y(y)}$$

The conditional pdf of Y given $X=x$ is

$$\underline{f_{Y|X=x}(y|x) = \frac{f(x,y)}{f_X(x)}}$$

Ex. Let (X, Y) has the joint PDF

$$f(x, y) = \begin{cases} 6xy(2-x-y), & 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

The marginal density function of the RV X is

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 6xy(2-x-y) dy \\ &= 6x \int_0^1 [(2-x)y - y^2] dy \\ &= 6x \left[(2-x) \frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 \\ &= 6x \left[(2-x) \frac{1}{2} - \frac{1}{3} \right] = (2-x)(3x) - 2x \end{aligned}$$

$$\therefore f_X(x) = \begin{cases} 4x - 3x^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \int_0^1 x(4x - 3x^2) dx = \frac{7}{12}$$

$$E(X^2) = \int_0^1 x^2(4x - 3x^2) dx = \frac{2}{5}$$

$$\therefore V(X) = E(X^2) - (E(X))^2 = \frac{2}{5} - \left(\frac{7}{12}\right)^2 = \frac{43}{720}$$

The marginal density of Y is $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$

$$\text{or, } f_Y(y) = \int_0^1 6xy(2-x-y) dx = 4y - 3y^2, \quad 0 < y < 1$$

$$E(Y) = \frac{7}{12}, \quad V(Y) = \frac{43}{720}.$$

covariance between the RVs X and Y is defined as

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

correlation coefficient between the RVs X and Y is

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}} = \frac{E(XY) - E(X)E(Y)}{\sqrt{V(X)V(Y)}},$$

$$-1 \leq \rho \leq 1$$

$$\rho = 0.8$$

$$\times \quad \rho = 0.8$$

$$\rho = -0.5$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) dx dy$$

$$= \int_0^1 \int_0^1 xy \cdot 6xy(2-x-y) dx dy$$

$$= \int_0^1 \int_0^1 6x^2y^2(2-x-y) dx dy$$

$$= \int_0^1 \left[\int_0^1 [6x^2(2-y)y^2 - 6x^3y^2] dx dy \right]$$

$$= \int_0^1 \left[2x^3(2-y)y^2 - \frac{3}{2}x^4y^2 \right]_{x=0}^1 dy$$

$$= \int_0^1 \left(2y^2(2-y) - \frac{3}{2}y^2 \right) dy$$

$$= \frac{1}{3} \cdot \checkmark$$

$$\text{Cov}(X, Y) = \frac{1}{3} - \frac{7}{12} \cdot \frac{7}{12} = -\frac{1}{144}, \quad \rho = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}} = \frac{-\frac{1}{144}}{\sqrt{\frac{43}{720} \cdot \frac{43}{720}}}$$

$$= -\frac{1}{144} \times \frac{720}{43}$$

$$\text{or } \boxed{\rho = -\frac{5}{43}}$$

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$$

$$= \frac{6xy(2-x-y)}{4y-3y^2}, \quad 0 < x < 1, \quad \underline{\underline{0 < y < 1}}$$

$$f_{X|Y=\frac{1}{2}}(x|\frac{1}{2}) = \frac{6x \cdot \frac{1}{2} (2-x-\frac{1}{2})}{4 \cdot \frac{1}{2} - 3(\frac{1}{2})^2}, \quad 0 < x < 1$$